

1. A new fiber with chromatic dispersion (β_2)

- $\beta_2 > 0$ at $\lambda = 1550 \text{ nm}$
- $\beta_2 < 0$ at $\lambda = 1310 \text{ nm}$

2. Optical sources have positive chirp ($k > 0$).

- β_2 (group velocity dispersion parameter)

- $\beta_2 > 0 \rightarrow$ normal dispersion $\rightarrow$ higher frequency components travel faster than lower frequency components

- $\beta_2 < 0 \rightarrow$ anomalous dispersion $\rightarrow$ lower frequency components faster than higher frequency components.

- Chirp (k): frequency modulation of the optical pulse
+ chirp $k > 0 \rightarrow$ instantaneous frequency increase with time.

Rule of thumb:

+ chirp + (+ β_2) $\rightarrow$ pulse broaden

+ chirp + (- β_2) $\rightarrow$ pulse can compress.

When we analyse for each wavelength

(i) $\lambda = 1550 \text{ nm}$.

- $\beta_2 > 0$ (normal dispersion)

- Source has $k > 0 \rightarrow$ pulse will broaden.

- Not ideal for high-speed

(ii) $\lambda = 1310 \text{ nm}$. (anomalous dispersion)

Source has $k > 0$ pulse may compress or broaden less
better for maintaining short pulses $\rightarrow$

Based on chromatic dispersion effects, we could use $\lambda = 1310$ nm since

- Neg β_2 are interacted by + chirp.
- This allows higher bit rates over longer distances.

2. Provided values:

Bit rate: 2.5 Gb/s

$$R_b = 2.5 \text{ Gb/s}$$

$$T_b = \frac{1}{R_b}$$

$$= \frac{1}{2.5 \times 10^9} \approx 0.4 \text{ ns}$$

Pulse width (i/p):

$$T_0 = 0.04 \text{ ns}$$

$$= 40 \text{ ps}$$

Wavelength: $\lambda = 1.55 \mu\text{m}$

Chirp: $k = 1.5$

Fiber dispersion parameter

$$D = 12 \text{ ps/nm}$$

$$\Delta\lambda = 0.5 \text{ nm}$$

Spectral width: $\Delta\lambda = 0.5 \text{ nm}$

$$\Delta T_{\text{disp}} = D \cdot \Delta \lambda \cdot L$$

L = fiber length (km)

$$\Delta T_{\text{disp}} = \left(12 \frac{\text{ps}}{\text{nm}} \right) (0.5 \text{ nm}) L = 6L \text{ ps.}$$

Chirp parameter:

$$\Delta T_{\text{total}} = (1 + K) \cdot D \cdot \Delta \lambda \cdot L$$

Substitute the values:

$$\Delta T_{\text{total}} = (1 + 1.5) (12) (0.5) L$$

$$\Delta T_{\text{total}} = (2.5) (6) L = 15L \text{ ps.}$$

for fiber length (max)

$$\Delta T_{\text{total}} \leq \frac{T_b}{2} = \frac{0.4}{2}$$

$$15L \leq 200$$

$$L \leq \frac{200}{15} \approx 13.3 \text{ km.}$$

The maximum fiber length is 13.3 km that can be used at 2.5 Gb/s.

3. Gaussian Pulse in dispersive medium:

$$T_z = T_0 \sqrt{1 + \left(\frac{4 \ln 2 \cdot \beta_2 \cdot L \cdot \Delta f}{T_0^2} \right)^2}$$

Wavelength spread $\Delta\lambda$ and dispersion parameters:

$$\Delta T = |\beta_2| \cdot L \cdot \Delta\omega$$

$\Delta\omega$ is spectral width

$$T_z = \sqrt{T_0^2 + (\beta_2 L \cdot \Delta\omega)^2}$$

1. Parameters given:

$$\Delta\lambda = 10 \text{ nm}$$

$$\Delta\lambda_{\text{ch}} = 1550 \text{ nm} = 1.55 \times 10^{-6} \text{ m}$$

$$\kappa = 0 \text{ (not chirped)}$$

$$\beta_2 = -20 \text{ ps}^2/\text{km}$$

$$T_0 = 0.5 \text{ ns} = 500 \text{ ps}$$

$$L = 10 \text{ km}$$

2. Convert $\Delta\lambda$ into $\Delta\omega$

$$\Delta\omega = \frac{2\pi c}{\lambda^2} \Delta\lambda$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\lambda = 1.55 \times 10^{-6} \text{ m}$$

$$\Delta\lambda = 10 \times 10^{-9} \text{ m}$$

$$\frac{c}{\lambda^2} = \frac{3 \times 10^8}{(1.55 \times 10^{-6})^2}$$

$$= \frac{3 \times 10^8}{2.4025 \times 10^{-12}} \approx 1.249 \times 10^{20}$$

$$\Delta \lambda = 1.0 \times 10^{-8}$$

$$1.249 \times 10^{20} \times 1.0 \times 10^{-8} = 1.249 \times 10^{12}$$

2π

$$\Delta \omega = 7.85 \times 10^{12} \text{ rad/s.}$$

Step

$$3. \quad \beta_2 L \cdot \Delta \omega = (20 \text{ ps}^2/\text{km}) (10 \text{ km}) (7.85 \times 10^{12} \text{ rad/s})$$

$$\Rightarrow 20 \text{ ps}^2/\text{km} = \frac{20 \cdot 10^{-24} \text{ s}^2}{1000 \text{ m}} = 20 \cdot 10^{-24} \text{ s}^2/\text{m} \times \frac{1}{1000 \text{ m.}}$$

$$= 20 \times 10^{-27} \text{ s}^2/\text{m.}$$

$$\Rightarrow 1 \text{ ps}^2/\text{km} = 20 \times 10^{-27} (10^{-12})^2 = 10^{-24} \text{ s}^2$$

Computing $20 \text{ ps}^2/\text{km}$

$$\Rightarrow 20 \text{ ps}^2/\text{km} = 20 \times 10^{-24} / 1000 \text{ s}^2/\text{m}$$

$$= 2.0 \times 10^{-26} \text{ s}^2/\text{m.}$$

$$L = 10 \text{ km} = 10^4 \text{ m.}$$

$$\beta_2 L = 2.0 \times 10^{-26} \times 10^4 = 2.0 \times 10^{-22} \text{ s}^2$$

$$\Delta \omega = 7.85 \times 10^{12}$$

$$\Rightarrow 2.0 \times 10^{-22} \times 7.85 \times 10^{12}$$

$$= 1.57 \times 10^{-9} = 1.57 \text{ ns}$$

Pulse width at z

$$T_z = \sqrt{T_0^2 + (1.57 \text{ ns})^2}$$

$$= \sqrt{(0.5 \text{ ns})^2 + (1.57 \text{ ns})^2}$$

$$= \sqrt{0.25 + 2.47} \approx \sqrt{2.72} \approx 1.65 \text{ ns.}$$

$$T_z = 1.65 \text{ ns.}$$

$$\text{Max bit rate} = \frac{1}{T_z}$$

$$R_b^{\text{max}} \approx \frac{1}{1.65 \text{ ns}} \approx 0.006 \text{ Gb/s}$$

$$R_b^{\text{max}} \approx 0.61 \text{ Gb/s}$$

If we compute for DSF with $\beta_2 = -5 \text{ ps}^2/\text{km}$

$$\text{New broadening} = 1.57 \times \frac{5}{20} = 0.392 \text{ ns.}$$

Substitute in

$$T_z = \sqrt{(0.5 \text{ ns})^2 + (0.392 \text{ ns})^2}$$

$$= \sqrt{0.25 + 0.154} = \sqrt{0.404}$$

$$T_z = 0.635 \text{ ns.}$$

$$\text{Max bit rate } R_b^{\text{max}} \approx \frac{1}{0.635 \text{ ns}} \approx 1.57 \text{ Gb/s}$$

for source $\Delta\lambda$ effect become negligible.

$$(\beta_2 L \Delta\omega)^2 \leq \frac{T_0^2}{10} \quad (1)$$

$$(\beta_2 L \Delta\omega)^2 \leq \frac{T_0^2}{\sqrt{10}} \quad (11)$$

~~Reason~~ ~~is~~ ~~rk~~

$$\beta_2 L \Delta\omega \leq \frac{T_0}{\sqrt{10}}$$

$$T_0 = 0.5 \text{ ns} = 0.5 \times 10^{-9} \text{ s}$$

$$\text{RHS} = 0.5 \times 10^{-9} / 3.162 \approx 1.58 \times 10^{-10} \text{ s} \rightarrow 0.158 \text{ ns}$$

$$\text{Current broadening} = 1.57 \text{ ns}$$

Now solve for $\Delta\lambda$ that satisfies.

$$\Delta\lambda_{\text{crit}} = \Delta\lambda \cdot \frac{0.158}{1.57}$$

$$\Delta\lambda_{\text{crit}} \approx 10 \text{ nm} \cdot 0.1006 \approx 1.0 \text{ nm}$$

~~Source bandwidth should be less than~~
Source bandwidth must be about $\Delta\lambda \leq 1 \text{ nm}$.
before its effect on dispersion becomes negligible.

4. Values provided:

(i) Transmitter power $\Rightarrow P_m = 1 \text{ mW} = 0 \text{ dBm}$.

(ii) Fiber length $L = 60 \text{ km}$.

(iii) Fiber attenuation $\alpha = 0.191 \text{ dB/km}$.

(iv) Splice every 6 km, Divide total fiber length by splice $\Rightarrow \text{Total splice} = \frac{60}{6} = 10$.

Fiber would be divided into 10 segments across 6 km each.

(v) Splice loss per splice: 0.25 dB .

(vi) Coupling loss: 6 dB .

(vii) Receiver sensitivity: $P_{Tn, \text{min}} = 20 \text{ pW}$.

Total loss calculation

(i) Fiber loss: $\alpha L = 0.191 \cdot 60 = 11.46 \text{ dB}$

(ii) Splice loss (dB) $10 \times 0.25 = 2.5 \text{ dB}$.

formula $\Rightarrow \text{Number of splices} \times \text{splice loss allowance}$.

(iii) Coupling loss: dB .

Total loss: Add all $\Rightarrow 11.46 + 2.5 + 6 = 19.96 \approx 20 \text{ dB}$

Power of fiber o/p.

$$\text{g/p power} = 0 \text{ dBm} = 1 \text{ mW}$$

$$P_{\text{out}} = P_{\text{in}} - \text{Total loss}$$

$$= 0 \text{ dBm} - 19.96 \text{ dBm} \approx -19.96 \text{ dBm}$$

$$\text{dBm } P_{\text{out}} = 10^{-\frac{19.96}{10}} \text{ mW}$$

$$= 10^{-1.996}$$

$$= 10 \times 1 \text{ mW} \approx 0.0101 \text{ mW}$$

$$P_{\text{w}} \approx 10.1 \mu\text{W}$$

We require $20 \mu\text{W}$ but we've got only $10 \mu\text{W}$ so the received power is low - the system will need amplifiers.

For Amplifiers.

We would need $10 \mu\text{W}$ for $20 \mu\text{W}$.

$$G \approx 10 \log_{10} \left(\frac{20 \mu\text{W}}{10 \mu\text{W}} \right)$$

$$= 10 \log_{10} (2 \mu\text{W})$$

$$G \approx 3 \text{ dB} \Rightarrow \text{amplifier needed.}$$

5. Given Parameters:

Fiber attenuation $\alpha = 0.18 \text{ dB/km}$

Center wavelength $\lambda_{\text{ch}} = 1555 \text{ nm}$

Chirp $x = 0$

Group velocity dispersion $\beta_2 = -20 \text{ ps}^2/\text{km}$

Input pulse width $T_0 = 0.1 \text{ ns}$

Bit rate $B = 1 \text{ Gb/s}$

Splice loss: 1 dB every 8 km

Coupling loss: 4 dB total

Transmitter bandwidth $\Delta\lambda = 0.5 \text{ nm}$

Tx power $P_{\text{in}} = 2 \text{ mW}$

Receiver sensitivity: $P_{\text{rx, min}} = 1 \mu\text{W}$

First we'll convert input power to dBm

$$P_{\text{in}} = 2 \text{ mW}$$

$$P_{\text{in}} (\text{dBm}) = 10 \log_{10} \left(\frac{2}{1} \right)$$

$$= 3.01 \text{ dBm}$$

Receiver sensitivity $P_{\text{rx, min}} = 1 \mu\text{W}$

$$= 10 \log_{10} \left(\frac{10^{-6}}{10^{-3}} \right) = -30 \text{ dBm}$$

$$\text{Link budget } 3 - (-30) = 33 \text{ dB}$$

loss calculation per km

$$\alpha = 0.18 \text{ dB/km}, \text{ splices } \rightarrow 8 \text{ km} \rightarrow 1 \text{ dB} \rightarrow 1/8 = 0.125 \text{ dB/km}$$

coupling loss 4 dB

$$\text{effective slope } \alpha_{\text{eff}} = 0.18 + 0.125$$
$$= 0.305 \text{ dB/km}$$

Max length by power budget.

$$\text{let length} = L$$

$$\text{Total loss} = 0.305L + 4$$

$$0.305L + 4 \leq 33$$

$$0.305L \leq 29$$

$$L \leq \frac{29}{0.305} \approx 95 \text{ km}$$

Chromatic dispersion broadening.

$$T_z = \sqrt{T_0^2 + \left(\frac{\beta_2 L \Delta \omega}{2T_0} \right)^2}$$

Unit conversion: $T_0 = 0.1 \text{ ns} = 1 \times 10^{-10} \text{ s}$

$$\beta_2 = -20 \text{ ps}^2/\text{km} = -20 \times 10^{-24} \text{ s}^2/\text{km}$$

$$\Delta \lambda = 0.5 \text{ nm at } \lambda = 1555 \text{ nm}$$

$\Delta\lambda$ to $\Delta\omega$.

$$\Delta\omega = \frac{2\pi c}{\lambda^2} \Delta\lambda$$

$$c = 3 \times 10^8 \text{ m/s}, \lambda = 1.553 \times 10^{-6} \text{ m}$$

$$\Delta\lambda = 0.5 \times 10^{-9} \text{ m.}$$

$$\Delta\omega = \frac{2\pi (3 \times 10^8) (0.5 \times 10^{-9})}{(1.553 \times 10^{-6})^2}$$

$$\Delta\omega = \frac{0.942}{2.418 \times 10^{-12}} \approx 3.9 \times 10^{11} \text{ rad/s}$$

Broadening.

$$\frac{\beta_2 L \Delta\omega}{2T_0} = \frac{(20 \times 10^{-24}) (3.9 \times 10^{11})}{2 \times 1 \times 10^{-10}}$$

$$= \frac{7.8 \times 10^{-12} L}{2 \times 10^{-10}} = 0.039 L$$

Total pulse limit width.

$$T_2 = \sqrt{(0.1)^2 + (0.039L)^2} \text{ ns.}$$

$$= \sqrt{0.01 + 0.001}$$

$$= 1 \text{ ns.}$$

dispersion limit

$$= \sqrt{0.1^2 + 0.039L^2} \leq 0.7$$

$$= 0.01 + (0.039L)^2 \leq 0.49$$

$$= (0.039L)^2 \leq 0.48$$

$$0.039L \leq 0.692$$

$$L \leq \frac{0.692}{0.039} \approx 17.7 \text{ km.}$$